

Volume II · Chapter 7 — The Hugging Face Ecosystem · Theory

Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-II-Chapter-7-context.md` for the AI interaction log.

Prerequisites: Ch 6.

Learning outcomes — the student can: navigate the Hub; use transformers pipelines and AutoModel/AutoTokenizer; load and process data with datasets; run efficient inference; manage models and licenses locally.

Topics (theory) — 8 h:

#	Topic	h
7.1	The Hub: models, datasets, spaces; model cards & licenses	1
7.2	transformers: pipelines, AutoModel/AutoTokenizer, config	2
7.3	datasets: loading, map, streaming	2
7.4	Tokenizers, batching, padding/truncation	1
7.5	Inference: CPU/GPU, half precision, device management	1
7.6	The evaluate library & metrics basics	1

Datasets/tools: HF transformers, datasets, evaluate; PubMedBERT and similar.

Assessment: inference-pipeline deliverable (50%); dataset-processing lab (30%); quiz (20%).

Key decisions: model choice/license; precision vs. memory; streaming vs. in-memory datasets.

References: plan §13 → *Transformers & the Hugging Face ecosystem*. **Hours:** Theory 8 + Lab 12 = 20.